

OPTICS

Interference of light:

If two beams of light cross each other at a certain point, in the region of cross over where both the beams are acting simultaneously, according to **superposition principle** a modification in their intensity is expected. The resultant intensity will be either great or less than that which would be given by one beam alone. This modification of intensity due to superposition of two or more beams of light is known as **interference of light**.

Superposition principle:

According to **Thomas Young**, when a medium is disturbed simultaneously by more than one wave, the instantaneous resultant displacement of the medium at every point at every instant is the algebraic sum of the displacement of the medium that would be produced at the point by the individual wave trains if each were present alone. After the superposition at the region of crossover, the wave trains emerge unimpeded as if they have not met each other at all. Each wave train retains its individual characteristics. Each wave train behaves as if others are absent.

Suppose two trains cross each other at a certain point and let y_1 be the displacement of the point produced by the first wave in the absence of second wave. If y_2 be the displacement of the point produced by the second wave in the absence of first wave, then the resultant displacement y of the point due to the two waves acting together is expressed by

$$y = y_1 + y_2 \dots\dots\dots (1)$$

If the two waves cross each other in phase, then Eq. (1) can be written as-

$$y = y_1 + y_2 \dots\dots\dots (2)$$

If the two waves cross each other out of phase, then Eq. (1) can be written as-

$$y = y_1 - y_2 \dots\dots\dots (3)$$

From equation (2) we can say that, if the two individual displacements are in the same direction, the resultant displacement will be enhanced. So, two waves reinforce each other and are said to produce constructive interference. For example, in [Fig-1.1(a)], two waves are of the same frequency but of different amplitudes, say a and b where $a > b$. When they reach a certain point in phase each other, then the resultant displacement or amplitude is equal to the sum of the two amplitude i.e., $(a + b)$.

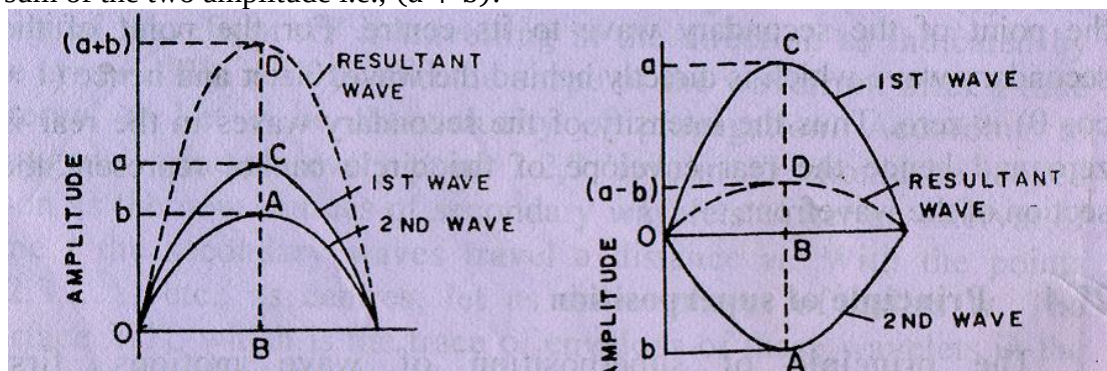


Fig.-1.1 (a) Constructive interference, (b) Destructive interference

From equation (3) we can say that, if the two individual displacements are in the opposite direction, the resultant displacement will be diminished. So, two waves neutralize each other and are said to produce **destructive interference**. For example, in Fig.-[1.1(b)], two waves are π or 180° out of phase with each other, the resultant amplitude is equal to the difference of the two amplitude i.e. $(a - b)$. If in addition, $a = b$, then the resultant amplitude is zero.

Conditions of interference:

- The two beams of light which interfere, must be coherent i.e., must originate from the same source of light.
- The two interfering waves must have the same amplitude.
- The original source must be monochromatic.
- The fringe-width should reasonably be as large as possible and the separation between the two sources should be as small as possible while the distance of the screen from the sources should be as large as possible.
- The two interfering waves must be propagated in almost the same direction.

Uses of interference:

- To determine the refractive index of liquids and gases.
- To determine the width of transparent materials.
- To determine the smoothness of a liquid and glass plate.
- To determine the wavelength of light.
- To measure the thickness of a very thin film.

Coherence Sources:

Two sources are said to be coherent if they emit light waves are always in phase with each other or with a constant phase difference.

In actual practice it is not possible to have two independent sources which are coherent. But for experimental purposes, two virtual sources formed from a single source can act as coherent sources.

Relation between path difference and phase difference:

If the path difference between the two waves is λ , the phase difference $= 2\pi$

Suppose for a path difference x , the phase difference is δ .

For a path difference λ , the phase difference $= 2\pi$

According to unitary method, we can write

The path difference λ is equal to phase difference 2π

$$\begin{array}{lcl} \therefore & 1 & \frac{2\pi}{\lambda} \\ \therefore & x & \frac{2\pi}{\lambda}x \end{array}$$

So, phase difference $\delta = \frac{2\pi}{\lambda}x$

Or, phase difference $(\delta) = \frac{2\pi}{\lambda} \times \text{path difference } (x)$

Or, $\frac{\delta}{2\pi} = \frac{x}{\lambda}$.

Interference of two light waves: analytical treatment

Consider a monochromatic source of light S emitting waves of wavelength λ and two slits A and B (Fig.-1.2). A and B are equivalent from S and act as two virtual sources. Let a be the amplitude of the waves. The phase difference between the two waves reaching the point P, at any instant, is δ .

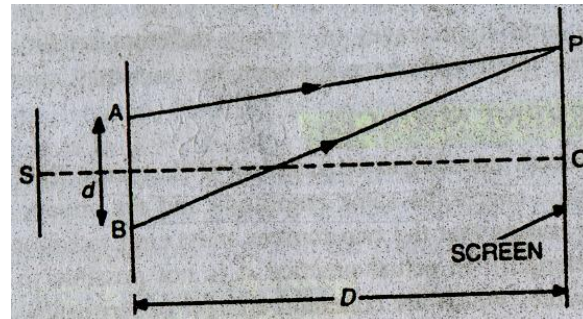


Fig.-1.2

Let y_1 be the displacement of the particle P due to waves emanating from the source A alone (in the absence of waves coming from B) at any instant of time.

$$y_1 = a \sin \omega t \dots\dots\dots (1)$$

Let y_2 be the displacement of the particle P due to waves emanating from the source B alone (in the absence of waves coming from A) at any instant of time.

$$y_2 = a \sin(\omega t + \delta) \dots\dots\dots (2)$$

If y be the resultant displacement of the particle P at that instant of time then according to principle of superposition-

$$\begin{aligned} y_1 + y_2 &= a \sin \omega t + a \sin(\omega t + \delta) \\ &= a \sin \omega t + a \sin \omega t \cos \delta + a \cos \omega t \sin \delta \\ &= a \sin \omega t (1 + \cos \delta) + a \cos \omega t \sin \delta \dots\dots\dots (3) \end{aligned}$$

$$\text{Taking } a(1 + \cos \delta) = R \cos \theta \dots\dots\dots (4)$$

$$\text{And } a \sin \delta = R \sin \theta \dots\dots\dots (5)$$

$$\begin{aligned} \therefore y &= R \sin \omega t \cos \theta + R \cos \omega t \sin \theta \\ &= R \sin(\omega t + \theta) \end{aligned}$$

which represents the equation of simple harmonic vibration of amplitude R. Squaring (4) and (5) and adding

$$\begin{aligned} R^2 \sin^2 \theta + R^2 \cos^2 \theta &= a^2 \sin^2 \delta + a^2 (1 + \cos \delta)^2 \\ R^2 &= a^2 \sin^2 \delta + a^2 + a^2 \cos^2 \delta + 2a^2 \cos \delta \\ R^2 &= 2a^2 + 2a^2 \cos \delta = 2a^2 (1 + \cos \delta) = 2a^2 \cdot 2 \cos^2 \frac{\delta}{2} \\ R^2 &= 4a^2 \cos^2 \frac{\delta}{2} \dots\dots\dots (6) \end{aligned}$$

We know that, the intensity at a point is given by the square of the amplitude

$$\therefore I = R^2$$

$$\text{Or, } I = R^2 = 4a^2 \cos^2 \frac{\delta}{2} \dots \dots \dots (7)$$

Special case:

(1) When the phase difference $\delta = 0, 2\pi, 2(2\pi), 2(3\pi), \dots \dots \dots n(2\pi)$ or the path difference $x = 0, \lambda, 2\lambda, \dots \dots \dots n\lambda$. The intensity is

$$I = 4a^2$$

Intensity is maximum when the phase difference is a whole number multiple of 2π or the path difference is a whole number multiple of wavelength.

(2) When the phase difference $\delta = \pi, 3\pi, \dots \dots \dots (2n+1)\pi$ or the path difference $x = \frac{\lambda}{2}, \frac{3\lambda}{2}, \dots \dots \dots \frac{(2n+1)\lambda}{2}$. The intensity is

$$I = 0$$

Intensity is minimum when the phase difference is a odd number multiple of π or the path difference is a odd number multiple of $\frac{\lambda}{2}$.

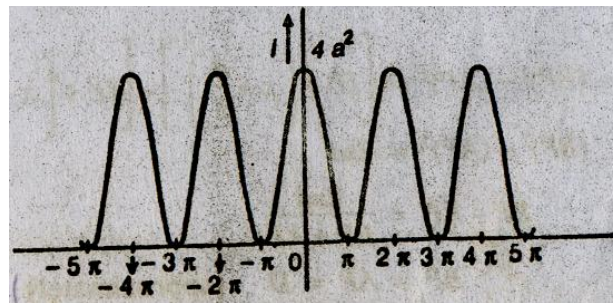


Fig.-1.3

From equation (7), it is found that the intensity at bright points is $4a^2$ and at dark points it is zero. According to the law of conservation of energy, the energy can not be destroyed. Here also the energy is not destroyed but only transferred from the points of minimum intensity to the points of maximum intensity.

Newton's rings:

When a plano-convex or bi-convex lens of large radius of curvature is placed on a glass plate P, a thin air film of progressively increasing thickness in all directions from the point of contact between the lens and the glass plate is very easily formed (Fig.-01). The air film thus possesses a radial symmetry about the point of contact. When it is illuminated normally with monochromatic light, an interference pattern consisting of a series of alternate dark and bright circular rings, concentric with the point of contact is observed (Fig.-02). The fringes are the loci of points of equal optical film thickness and gradually become narrower as their radii increase until the eye or the magnifying instrument can no longer separate them. The rings are localized in the air film. Since the phenomenon was first examined in detail by Newton, the rings are termed as Newton's rings.

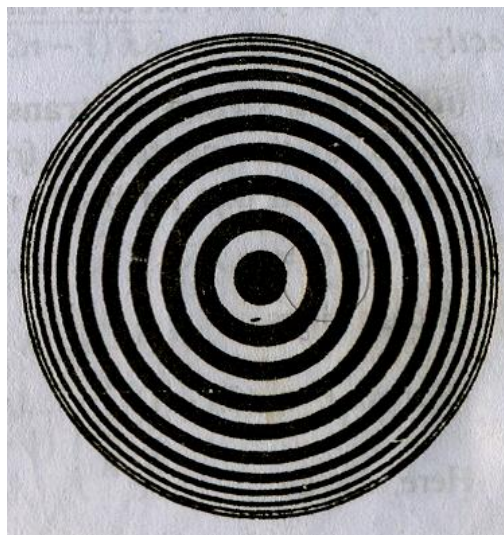


Fig.-1.8: Newton's Rings

Diffraction of Light:

When light from a narrow linear slit is incident on the sharp edge of an obstacle, it will be found that there is illumination to some extent within the geometrical shadow of the obstacle. This shows that light can be bent round an obstacle and this bending of light round an obstacle is called diffraction of light.

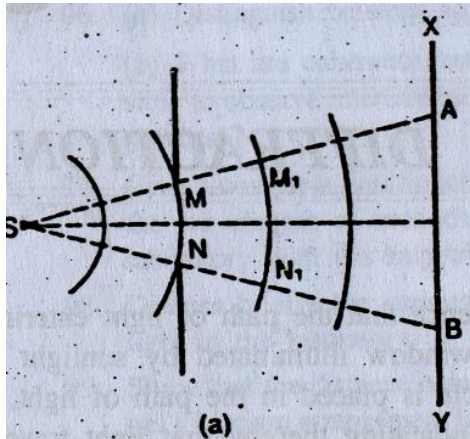


Fig.-1.1(a)

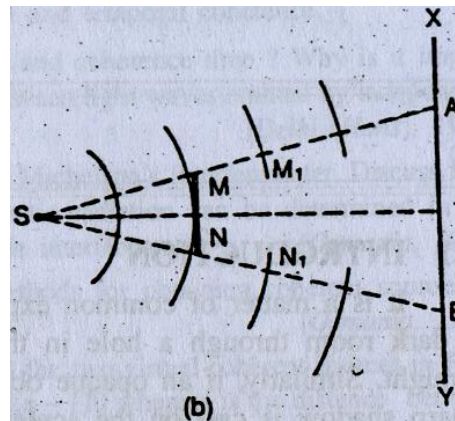


Fig.-1.1 (b)

Explanation:

Let, S is a source of mono-chromatic light and MN is a small aperture, XY is the screen placed in the path of light. AB is the illuminated portion of the screen and above A and below B is the region of the geometrical shadow.

Considering MN as the primary wave-front, according to the Huygen's construction, if the secondary wavefronts are drawn the encroachment of light in the geometrical shadow is expected. This bending of light round the edge of an obstacle or the encroachment of light in the geometrical shadow is called diffraction (Fig.-1). Similarly, if an obstacle MN is placed in the path of light (Fig.-2), there should be illumination in the geometrical shadow region AB also.

Classification of the diffraction.

Diffraction phenomena can be divided into two groups. These are following:

1. Fresnel diffraction phenomena
2. Fraunhofer diffraction phenomena

1. Fresnel diffraction phenomena:

In the Fresnel class of diffraction, the source or the screen or both are at finite distances from the aperture or obstacle causing diffraction.

2. Fraunhofer diffraction phenomena:

In the Fraunhofer class of diffraction phenomena, the source and the screen on which the pattern is observed are at infinite distances from the aperture or the obstacle causing diffraction.

Distinction between Fresnel and Fraunhofer diffraction:

The difference between Fresnel and Fraunhofer diffraction is given below:

Fresnel diffraction	Fraunhofer diffraction
1. The source or the screen or both are at finite distances from the aperture.	1. The source and the screen are at infinite distance from the aperture.
2. Observation of Fresnel diffraction phenomena does not require any lenses.	2. The incoming light is rendered parallel with a lens and the diffracted beam focused on the screen with another lens.
3. Theoretical treatment is complex.	3. Theoretical treatment is simpler.
4. The wave-fronts are spherical and hence divergent.	4. The wave-fronts are planer.
5. It is used to construct a zone plate.	5. It is used to construct plane transmission grating or concave reflection grating.
6. Spectral lines that are seen are less brighter.	6. Spectral lines that are seen are more brighter.
7. Centre of the screen can be both bright and dark. It depends upon the width of the slit.	7. Centre of the screen is always bright.

Intensity pattern due to single slit diffraction The maxima and minima of the pattern.

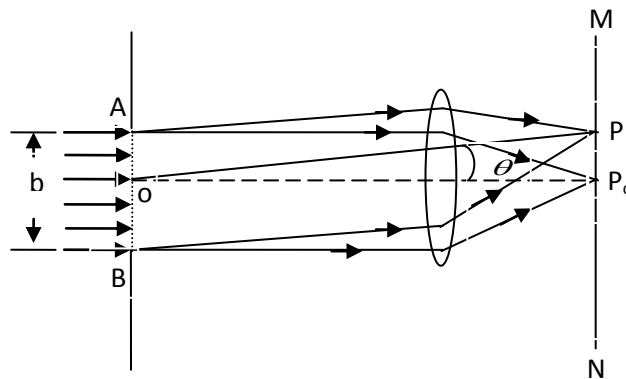


Fig.-1.2: Fraunhofer diffraction by a single slit

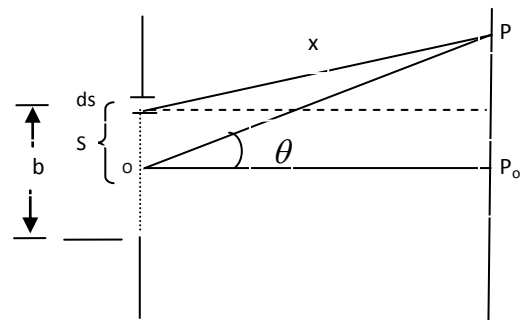
Let a parallel beam of mono-chromatic light of wavelength λ be incident normally on a narrow slit AB of width b through which parallel rays are passing. Let ds be the small element at a distance s from the origin O.

The part of each secondary wave which travels normal to the plane of the slit will be focused at P_0 , while those which travel at any angle θ will reach P. Considering first the wave-length emitted by the element ds situated at the origin. The amplitude will be proportional to the length ds and inversely proportional to the distance, so

$$dy_0 = \frac{ads}{x} \sin (wt - kx) \dots \dots \dots (1)$$

Where, a = amplitude of wave

$$k = \frac{2\pi}{\lambda}$$



As the position ds varies the displacement it produces will vary in phase because of the different path length to P. When ds is situated at a distance S from the origin O, then,

$dy_s = \frac{ads}{x} \sin [wt - k(x + \Delta x)]$ $= \frac{ads}{x} \sin [wt - k(x + S \sin \theta)]$ $= \frac{ads}{x} \sin [wt - kx - kS \sin \theta] \dots \dots \dots (2)$	$\Delta x = S \sin \theta$ = Phasedifference	
---	---	--

Similarly,

$$dy_{-s} = \frac{ads}{x} \sin [wt - kx + kS \sin \theta] \dots\dots\dots (3)$$

We know,

$$\begin{aligned} dy &= dy_s + dy_{-s} \\ &= \frac{ads}{x} \sin [wt - kx - kS \sin \theta] + \frac{ads}{x} \sin [wt - kx + kS \sin \theta] \\ &= \frac{ads}{x} [\sin (wt - kx - kS \sin \theta) + \sin (wt - kx + kS \sin \theta)] \\ &= \frac{ads}{x} [2 \cos(kS \sin \theta) \sin (wt - kx)] \dots\dots\dots (4) \end{aligned}$$

Now, integrating equation (4) between the limits 0 to b/2 and x is constant, we get

$$\begin{aligned} y &= \frac{2a}{x} \sin (wt - kx) \int_0^{b/2} \cos(kS \sin \theta) ds \\ y &= \frac{2a}{x} \sin (wt - kx) \left[\frac{\sin(kS \sin \theta)}{k \sin \theta} \right]_0^{b/2} \\ &= \frac{2a}{x} \left[\frac{\sin\left(\frac{1}{2} kbs \sin \theta\right)}{k \sin \theta} \right] \sin (wt - kx) \\ y &= \frac{ab}{x} \frac{\sin\left(\frac{1}{2} kbs \sin \theta\right)}{\frac{1}{2} kbs \sin \theta} \sin (wt - kx) \dots\dots\dots (5) \end{aligned}$$

The resultant vibration will therefore be a simple harmonic one

We can write equation (5) as,

$$y = A \sin (wt - kx)$$

Where, amplitude $A = \frac{ab}{x} \frac{\sin\left(\frac{1}{2} kbs \sin \theta\right)}{\frac{1}{2} kbs \sin \theta}$

$$= \frac{ab}{x} \frac{\sin \beta}{\beta} = A_0 \frac{\sin \beta}{\beta} \dots\dots\dots (6)$$

We know, the intensity is square of the amplitude, so intensity,

$$I = A^2$$

$$= A_0^2 \frac{\sin^2 \beta}{\beta^2} \dots\dots\dots (7)$$

Which is the required expression.

From (7)

$$I = A_0^2 \frac{\sin^2 \beta}{\beta^2} = I_0 \frac{\sin^2 \beta}{\beta^2}$$

Differentiating,

$$dI = I_0 \left[\frac{\beta^2 2 \sin \beta \cos \beta - (\sin^2 \beta) 2\beta}{\beta^4} \right] d\beta$$

For I to be maximum, $\frac{dI}{d\beta} = 0$

$$\beta^2 (2 \sin \beta \cos \beta) - (\sin^2 \beta) 2\beta = 0$$

$$\text{or, } \beta^2 (2 \sin \beta \cos \beta) = (\sin^2 \beta) 2\beta$$

$\therefore \tan \beta = \beta$; transcendental equation.

The maxima and minima of the pattern

When $\beta = 0$, then $\lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1$

$$I = A_0^2$$

Which corresponds to maximum intensity. The intensity is zero when $\beta = m\pi$ and $m \neq 0$.

Substituting the value of β , we get

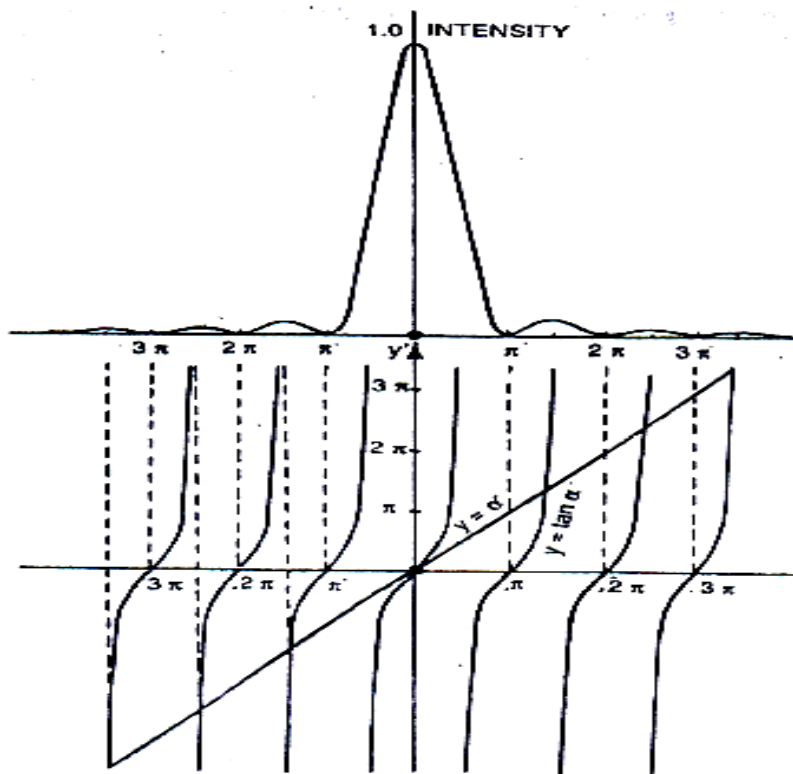
$$\beta = \frac{1}{2} kb \sin \theta$$

$$m\pi = \frac{1}{2} kb \sin \theta$$

$$m\pi = \frac{1}{2} \frac{2\pi}{\lambda} b \sin \theta \quad \left[\because k = \frac{2\pi}{\lambda} \right]$$

$$b \sin \theta = m\lambda \dots\dots\dots (8)$$

Here $m = 1, 2, 3, \dots\dots\dots$, etc and we will get minimum intensity distribution.



The position of secondary maxima are obtained by putting $\beta = \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \dots\dots\dots etc.$

From equation (8) we see that when the breadth b is wide $\sin \theta$ will be small. Then the minimum and maximum intensity will approach to the central maximum intensity at P_0 . The value of θ will be large when the slit breadth is small. In this case the maximum and maximum diffraction will be observed on both sides of the principle maximum.

Diffraction grating:

Any arrangement which is equivalent in its action to a number of parallel equidistant slits of the same width is called a diffraction grating. It is an extremely useful device. There are a large number of narrow slits side by side. The slits are separated by opaque space.

There are three kinds of diffraction grating:

1. Plane diffraction grating
2. Concave reflecting grating and
3. Echelon grating.

Polarization of light:

Ordinary light consists of transverse vibrations, that is, the vibrations are at right angles to the direction of propagation of the wave. An ordinary beam of light consists millions of such waves, each with its own plane of vibration. If, by some means the vibrations constituting the beam of light are confined to one plane, the light is said to be plane polarized.

The process by which light vibrations are confined to one particular direction is known as polarization.

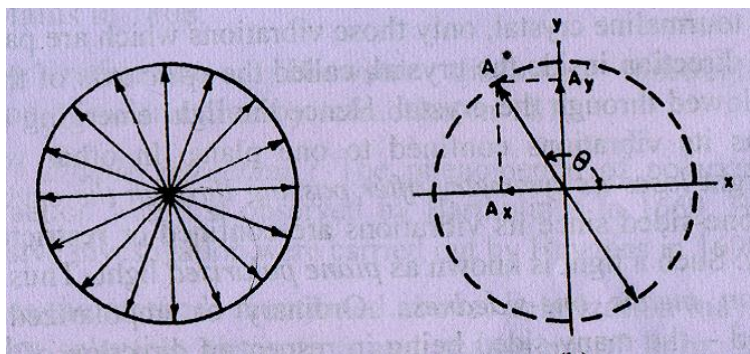


Fig.-1.1: Polarization of light

By properly constructed nicol prism of tourmaline crystal and quarter wave plate, polarized light can be obtained. Also, polarized light can be obtained. Also; polarized light can be produced by reflection, refraction, and double-refraction. Polarized light are of three types, that are circularly polarized, elliptically polarized and plane polarized.

From the properties of interference and diffraction we are led to conclude that light is a wave phenomenon and we utilize these properties to measure the wavelength. These effects tell us nothing about the type of waves with which we are dealing- whether they are longitudinal or transverse, or whether the vibrations are linear, circular, or torsional. The phenomenon of polarization has helped to establish beyond doubt that light waves are transverse waves.

Polarization of transverse wave:

Let light from a source S fall on a tourmaline crystal A which is cut parallel to its axis (fig.-1). The crystal A will act as the slit S1. On rotating the crystal A, no remarkable change is noticed. Now place the crystal B parallel to A.

- (i) Rotate both the crystals together so that their axes are always parallel. No change is observed in the light coming out of B.
- (ii) Keep the crystal A fixed and rotate the crystal B. The light transmitted through B becomes dimmer and dimmer. When B is at right angles to A, no light emerges out of B (fig.-1).

If the crystal B is further rotated, the intensity of light coming out of it gradually increases and is maximum again when the two crystals are parallel.

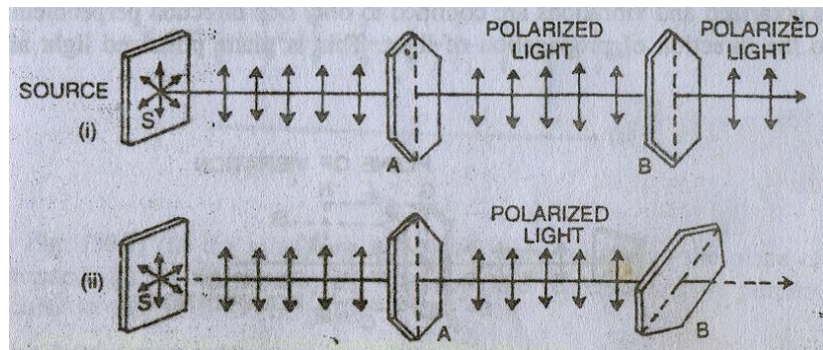


Fig.-1.2: Light waves are transverse waves

This experiment shows conclusively that light is not propagated as longitudinal or compressional waves. If we consider the propagation of light as a longitudinal wave motion then no extinction of light should occur when the crystal B is rotated. It is clear that after passing through the crystal A, the light waves vibrate only in one direction. Therefore light coming out of the crystal A is said to be polarized because it has acquired the property of one sidedness with regard to the direction of the rays.

This experiment proves that light waves are transverse waves, otherwise light coming out of B could never be extinguished by simple rotating the crystal B.

Plane of polarization and vibration:

Ordinary or unpolarized light is many-sided-the many-sided being in respect of direction of vibration. When ordinary light is passed through a tourmaline crystal, the light is polarized and vibrations are confined to only one direction perpendicular to the direction of propagation of light. This is plane polarized light and it has acquired the property of sidedness.

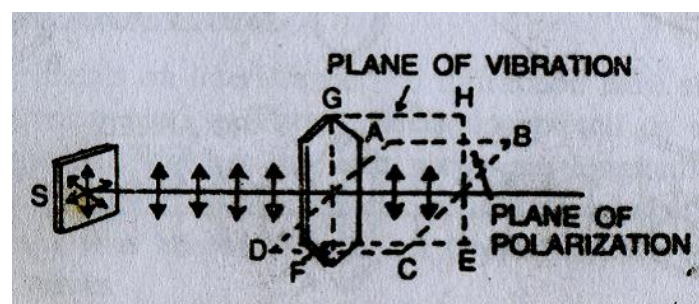


Fig.-1.3: Plane of polarization and Plane of vibration

The plane in which vibrations take place is known as the plane of vibration. The plane in which no vibrations occur is the plane of polarization and is obviously at right angles to the plane of vibration. The plane ABCD in fig.-2 is the plane of polarization. The EFGH in fig.-2 is the plane of vibration.

Methods of producing plane-polarized light:

There are three important methods for producing plane-polarized light. these are:

- (i) by reflection: Polarization of light by reflection was first noted by Malus in 1808.
- (ii) by refraction: Brewster discussed thoroughly the phenomenon of polarization by refraction in 1812.
- (iii) by double refraction: The phenomenon of polarization by double refraction was first observed by Bartholinus in 1669. However, comprehensive investigation were carried out by Huygens in 1690.

Besides the methods mentioned above, polarization may also be produced by selective absorption and scattering of light.

Polarization by reflection:

Polarization of light by reflection from the surface of glass was first discovered by Etienne Louis Malus in 1808. He observed that when a beam of natural light is incident on a glass surface CD along the path AB, both the reflected light along BE and the transmitted light along BM are partially polarized (fig.-3). The reflected light consists mostly of dot components along with a few arrow components. The transmission light mostly consists of arrow components along with a good number of dot components. This is true for all angles of incidence except one particular angle. At this particular angle of incident, none of the arrow components is reflected; they are all transmitted. Of the dot components about 15% are reflected in the case of the glass surface (The fraction refracted depends upon the refractive index of the reflecting surface), the rest being transmitted along with the arrow components. The reflected light, although weak, is completely polarized (fig.-2 (b)).

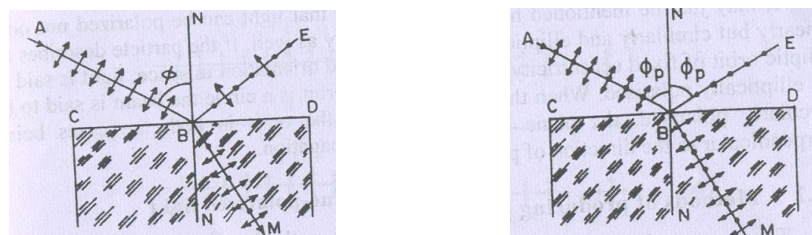


Fig.-1.4: Polarization by reflection and refraction.

On the other hand, refracted light is strong but only partially polarized. This particular angle of incidence at which the reflected light becomes completely polarized is known as the polarized angle and is equal to 57° for a glass surface. At any other angle of incidence, the reflected light will not be completely polarized and will always contain a certain amount of arrow components.

Polarization by reflection or Brewster's law:

Sir David Brewster, a Scottish Physicist, found experimentally in 1812 that the polarizing angle depends upon the refractive indices of the refracting material and the surrounding medium in which it is placed. According to Brewster, the tangent of the polarizing angle is equal to the refractive index of the refracting material with respect to its surroundings. This is known as Brewster's law.

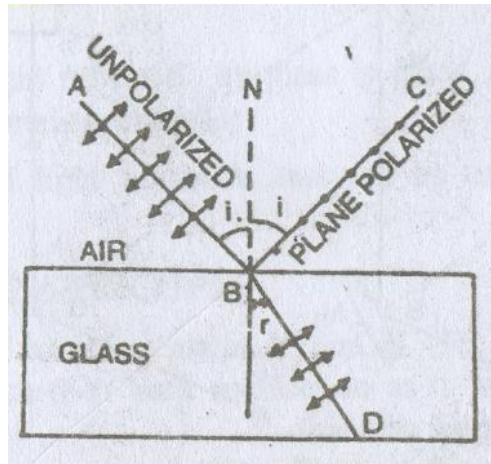


Fig.-1.5: Brewster's law for the polarizing angle.

Statement:

When the angle of incidence is equal to the angle of polarization for a transparent substance, the tangent of the angle of polarization will be equal to the refractive index (μ) of the substance. Moreover, the reflected and the refracted rays are perpendicular to each other.

If ϕ – be polarizing angle and μ be the refractive index of the medium, then according to the Brewster's law,

$$\mu = \tan \phi$$

Explanation:

Let an unpolarized light is incident at an angle equal to the polarizing angle on the glass – surface. It is reflected along BC and refracted along BD. From Snell's law,

$$\mu_o = \frac{\sin i}{\sin r} = \frac{\sin \phi}{\sin r} \dots \dots \dots (1) \quad [\phi = i]$$

From Brewster's law

$$\begin{aligned} \mu &= \tan \phi \\ &= \frac{\sin \phi}{\cos \phi} \dots \dots \dots (2) \end{aligned}$$

Comparing equation (1) and (2), we get

$$\Rightarrow \frac{\sin \phi}{\cos \phi} = \frac{\sin \phi}{\sin r}$$

$$\Rightarrow \cos \phi = \sin r$$

$$\Rightarrow \cos \phi = \cos \left(\frac{\pi}{2} - r \right)$$

$$\Rightarrow \phi = \frac{\pi}{2} - r$$

$$\Rightarrow \phi + r = \frac{\pi}{2} \dots\dots\dots (3)$$

As $\phi + r = \frac{\pi}{2}$, $\angle CBD$ is also equal to $\frac{\pi}{2}$. Therefore, the reflected and the refracted rays are at right angles to each other.

It is not difficult to understand the physical reason why the light vibrating in the plane of incidence is not reflected at Brewster's angle. The incident light sets the electrons in the atoms of the material into oscillation, and it is the re-radiation from these that generates the reflected beam. When the latter is observed at 90° to the refracted beam, only the vibrations that are perpendicular to the plane of incidence can contribute. Those in the plane of incidence have no component traverse to the 90° direction and hence cannot radiate in that direction. The reason is the same as that which causes the radiation from a horizontal radio-transmitter antenna to drop to zero along the direction of the wires.

Intensity of polarized light or Malus' Law:

We have seen before that light, polarized by a tourmaline crystal, will pass through a second tourmaline crystal if the optic axis of this second crystal is parallel to that of the first crystal. This first crystal is referred to as the polarizer while the second crystal is known as the analyzer as it helps to detect polarized light since the eye alone cannot detect if light is polarized. When the optic axes of both the polarizer and the analyzer are parallel to each other, the intensity of the emergent polarized light is maximum. Let this maximum intensity be I_0 . As the analyzer is gradually rotated, the intensity of the transmitted light goes on decreasing until it becomes zero, when the optic axes of the two crystals are at right angles to each other. If the rotation of the analyzer is continued, then light gradually re-appears until the original intensity of the emergent light is restored when the analyzer has turned through 180° with respect to the polarizer i.e., the two optic axes again become parallel to one another. Consider the case when the analyzer has been rotated through an angle θ from its position of maximum intensity. The amplitude A of the plane polarized light incident on the analyzer can be resolved into two components perpendicular to each other. The component $A\cos\theta$ is transmitted while the component $A\sin\theta$ is either rejected or absorbed. Since the intensity is proportional to the square of the amplitude, the intensity I of the transmitted light at this position is given by

$$I \propto A^2 \cos^2 \theta$$

$$\text{and also } I \propto A^2$$

$$I = I_0 \cos^2 \theta ; \text{ or, } \frac{I}{I_0} = \cos^2 \theta$$

This is known as Malus's cosine law according to which the intensity of the polarized light emerging from, the analyzer is proportional to the square of the cosine of the angle between the polarizer and the analyzer.

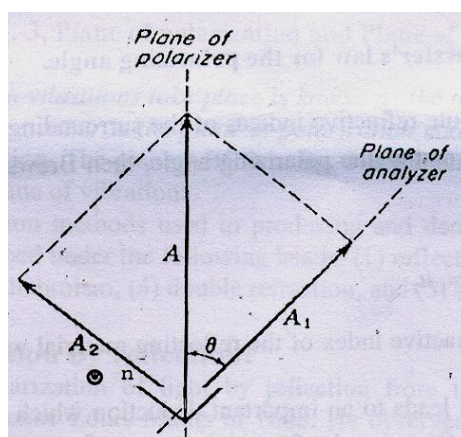


Fig.-1.6:Malus' Law